

Antenna Datasheet

Product OC: YECT001W1EM
Version: 1.0
Date: 2026-07-08
Status: Released

Product Name: 4G Terminal Mount Rubber Monopole External Antenna
Key Features:
Frequency Band: 698–960 MHz, 1710–2690 MHz
Dimensions: 52.6 mm × 18.6 mm × 9 mm
Efficiency: Up to 78.9 % (EVB)
RoHS & REACH Compliant
IP54

Overview

YECT001W1EM is a 4G rubber antenna measuring 52.6 mm × 18.6 mm × 9 mm. This ultra-wide-band 4G antenna provides broad coverage from 698–960 MHz, 1710–2690 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is terminated with 90° SMA Male connector. This low profile, terminal mount omni-directional antenna, ideal for applications where the antenna is required to be discrete, is easy to install with maximum durability assured thanks to its TPE enclosure. It is compatible with Quectel ‘s 4G Series modules.

It allows constant and reliable transmission and reception due to its omni-directional gain across all frequency bands. YECT001W1EM is designed as a monopole antenna, which needs to be mounted on a ground plane to offer high efficiency in all working bands. It is a perfect antenna product for customers that desire highest performance. This high-efficiency, high-gain omni-directional antenna is ideally suited for Zigbee, Bluetooth, IoT Sensors, public safety and security, point of sales terminals, smart home automation, robotics / autonomous.

- **Typical applications include:**

✓ Zigbee
✓ Bluetooth
✓ IoT Sensors
✓ Public Safety and Security
✓ Point of Sales Terminals
✓ Smart Home Automation
✓ Robotics / Autonomous

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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1 Specification

Test Condition: Free Space & On 130 mm × 130 mm EVB

1.1. Electrical

Electrical	
Frequency Range	698–960 MHz, 1710–2690 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional
Antenna Type	Monopole

Electrical – Detail											
Band / SPEC	Band	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR	FS	9.8	3.4	-	9.3	4.0	5.7	8.8	-	-	-
	EVB	5.4	7.1	-	2.9	1.7	1.6	4.8	-	-	-
Max. Return Loss (dB)	FS	-1.8	-5.3	-	-1.9	-4.4	-3.1	-2.0	-	-	-
	EVB	-3.2	-2.5	-	-6.3	-11.6	-12.7	-3.7	-	-	-
AVG Eff. (%)	FS	19.1	27.7	-	37.3	33.3	26.5	20.0	-	-	-
	EVB	54.9	47.8	-	57.1	75.9	69.1	50.0			
AVG AVG Gain (dB)	FS	-7.3	-5.6	-	-4.5	-4.8	-5.8	-7.0	-	-	-
	EVB	-2.7	-3.4	-	-2.4	-1.2	-1.6	-3.1	-	-	-
Max. Peak Gain (dBi)	FS	-2.8 (810)	-1.5 (940)	-	1.0 (2030)	-1.0 (2310)	-2.1 (2400)	-2.8 (2500)	-	-	-
	EVB	1.0 (810)	1.1 (820)	-	2.3 (2170)	2.6 (2300)	1.5 (2400)	-0.1 (2500)	-	-	-
VSWR	FS	≤ 9.8									
	EVB	≤ 7.1									
Return Loss	FS	≤ -1.8 dB									
	EVB	≤ -2.5 dB									
Peak Gain	FS	≤ 1.0 dBi									
	EVB	≤ 2.6 dBi									

1.2. Supported Bands

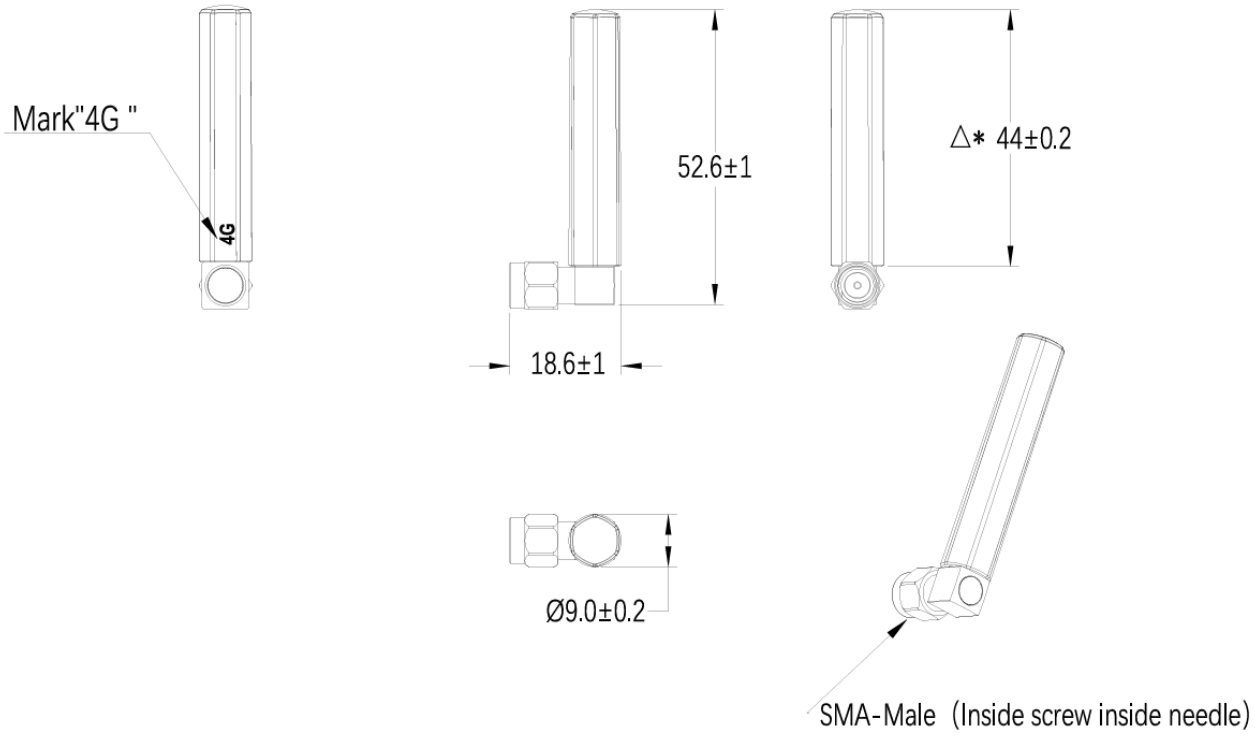
5G NR / LTE / LTE-Advanced / WCDMA / HSPA / HSPA+ / GPRS / GSM / NB-IoT				
Band	**Frequency (MHz)**	**Uplink (MHz)**	**Downlink (MHz)**	**Covered**
1	2100	1920–1980	2110–2170	√
2	1900	1850–1910	1930–1990	√
3	1800	1710–1785	1805–1880	√
4	1700	1710–1755	2110–2155	√
5	850	824–849	869–894	√
7	2600	2500–2570	2620–2690	√
8	900	880–915	925–960	√
9	1800	1749.9–1784.9	1844.9–1879.9	√
11	1500	1427.9–1447.9	1475.9–1495.9	-
12	700	699–716	729–746	√
13	700	777–787	746–756	√
14	700	788–798	758–768	√
17	700	704–716	734–746	√
18	850	815–830	860–875	√
19	850	830–845	875–890	√
20	800	832–862	791–821	√
21	1500	1447.9–1462.9	1495.9–1510.9	-
22	3500	3410–3490	3510–3590	-
23	2100	2000–2020	2180–2200	√
24	1600	1626.5–1660.5	1525–1559	-
25	1900	1850–1915	1930–1995	√
26	850	814–849	859–894	√

28	700	703–748	758–803	√
31	450	452.5–457.5	462.5–467.5	-
34	2100	2010–2025		√
38	2600	2570–2620		√
39	1900	1880–1920		√
40	2300	2300–2400		√
41	2500	2496–2690		√
42	3500	3400–3600		-
48	3500	3550–3700		-
66	1700	1710–1780	2110–2200	√
71	600	663–698	617–652	-
74	1500	1427–1470	1475–1518	-
77	3500	3300–4200		-
78	3500	3300–3800		-
79	4500	4400–5000		-

1.3. Mechanical & Environmental

Mechanical	
Antenna Dimensions	52.6 mm × 18.6 mm × 9 mm
Casing Material & Color	TPE & Black
Connector Type	90° SMA Male
Mounting Type	Terminal
Weight	Typ. 7.4 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
Ingress Protection (IP) Rating	IP54
RoHS & REACH Compliant	Yes

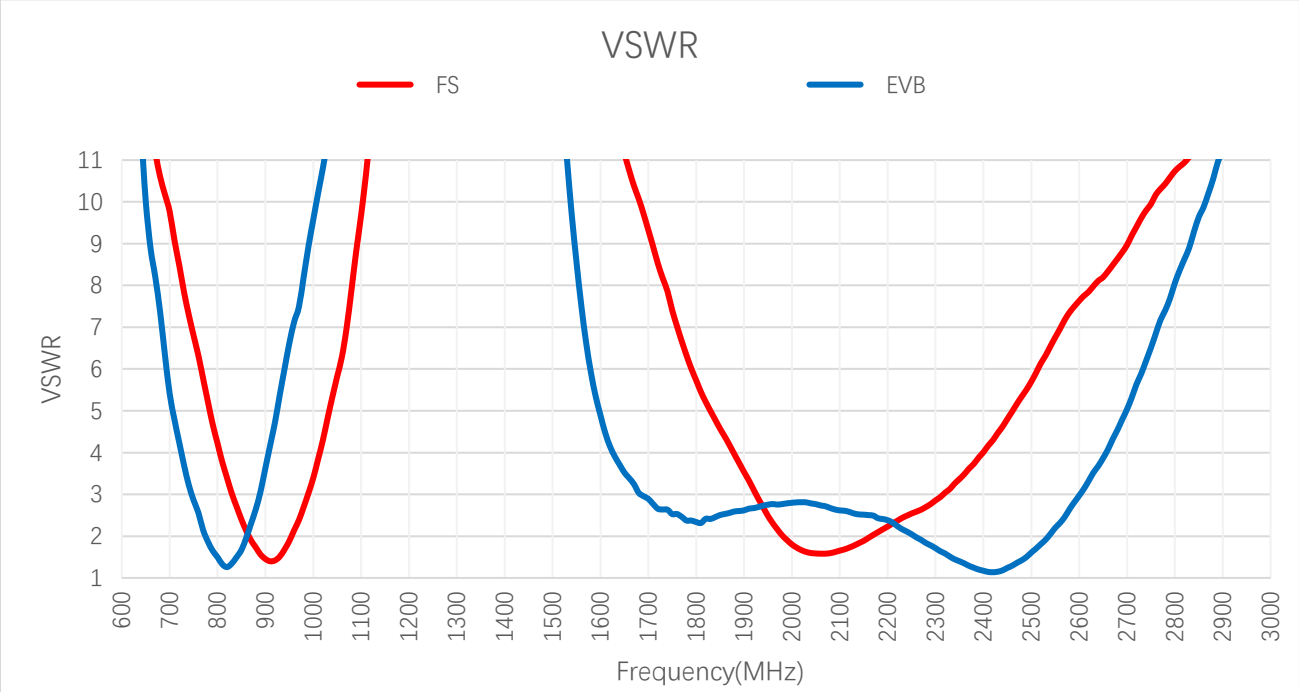
2 Drawing


Mark"4G "
4G
52.6±1
△* 44±0.2
18.6±1
Ø9.0±0.2
SMA-Male (Inside screw inside needle)

3 Detailed Performance

3.1. S-Parameter Test

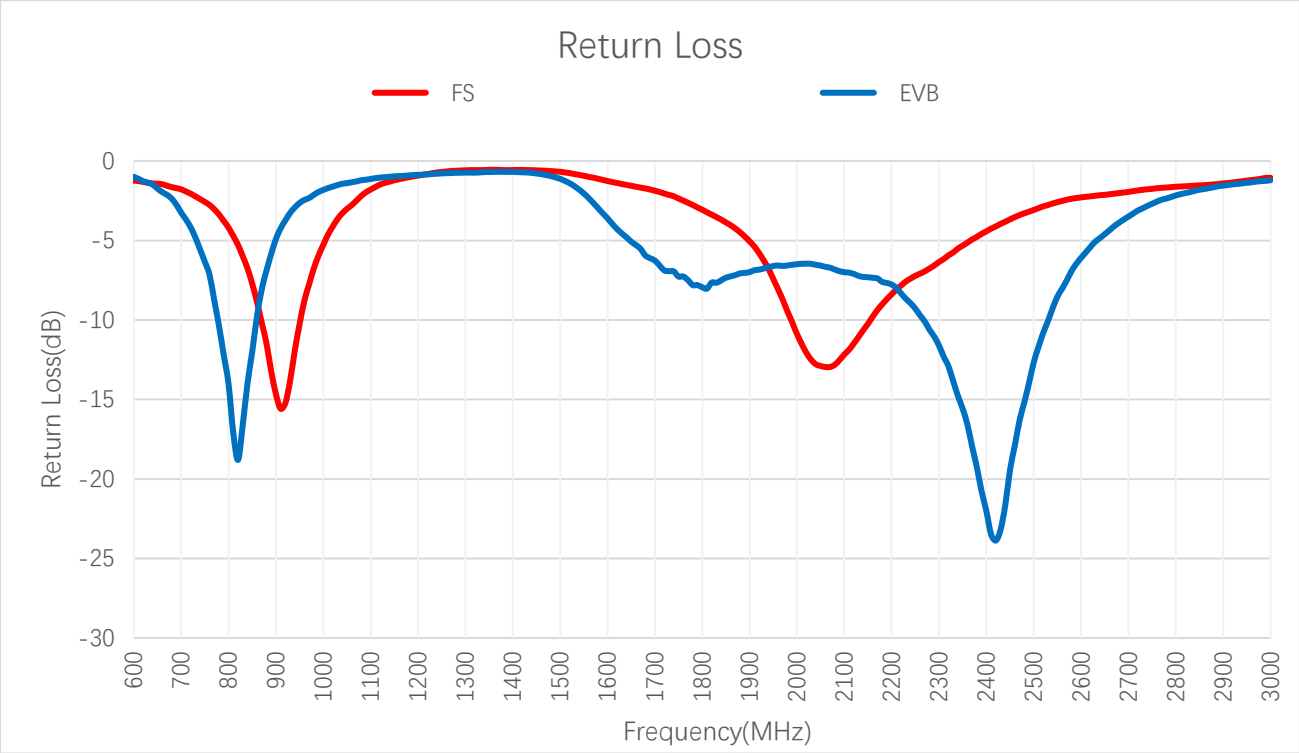
3.1.1. VSWR



VSWR

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	9.1	3.0	1.5	2.1	-	8.9	7.9	4.0
EVB	-	-	4.8	1.3	3.6	7.1	-	2.8	2.6	2.6
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	2.5	1.8	3.4	4.8	7.6	8.8	-	-	-	-
EVB	2.8	2.5	1.4	1.2	3.0	4.8	-	-	-	-

3.1.2. Return Loss

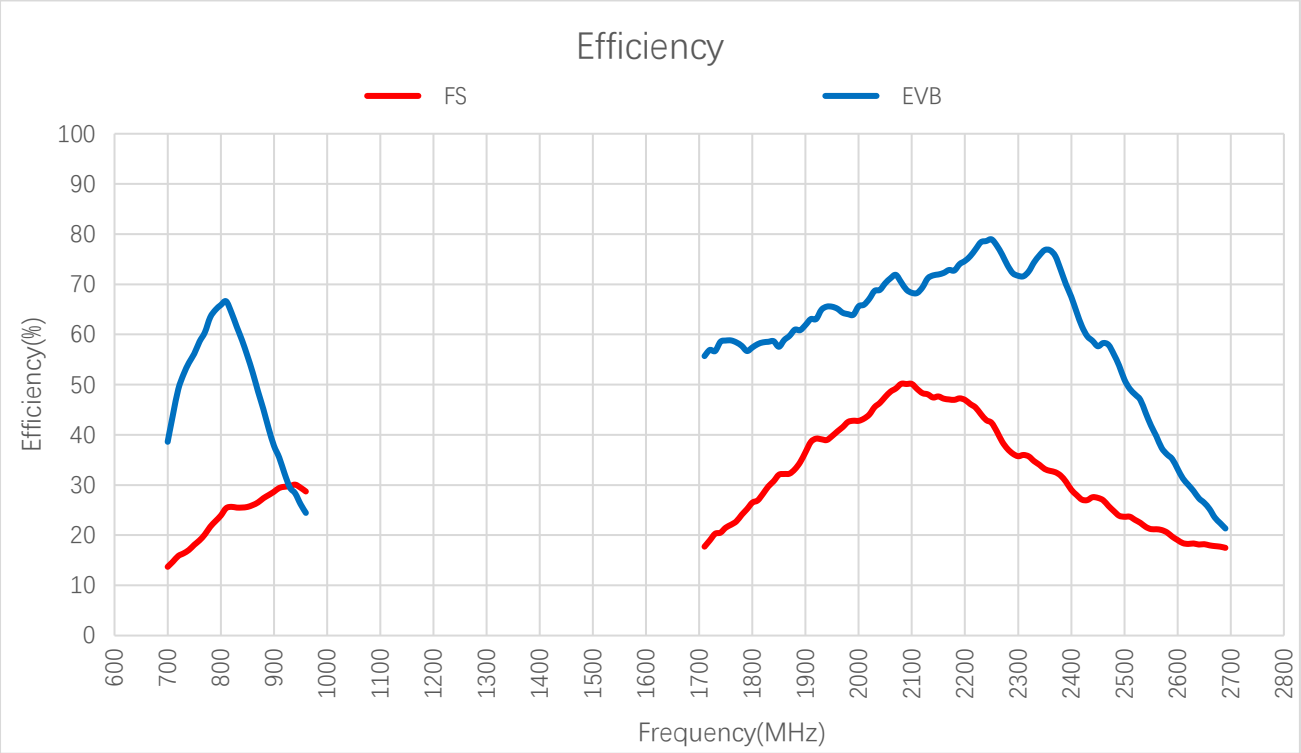


Return Loss (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-1.9	-6.0	-14.7	-8.8	-	-2.0	-2.2	-4.5
EVB	-	-	-3.7	-16.6	-4.9	-2.5	-	-6.6	-6.9	-7.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-7.4	-10.6	-5.3	-3.7	-2.3	-2.0	-	-	-	-
EVB	-6.6	-7.3	-15.6	-19.4	-6.1	-3.7	-	-	-	-

3.2. Radiation Performance Test

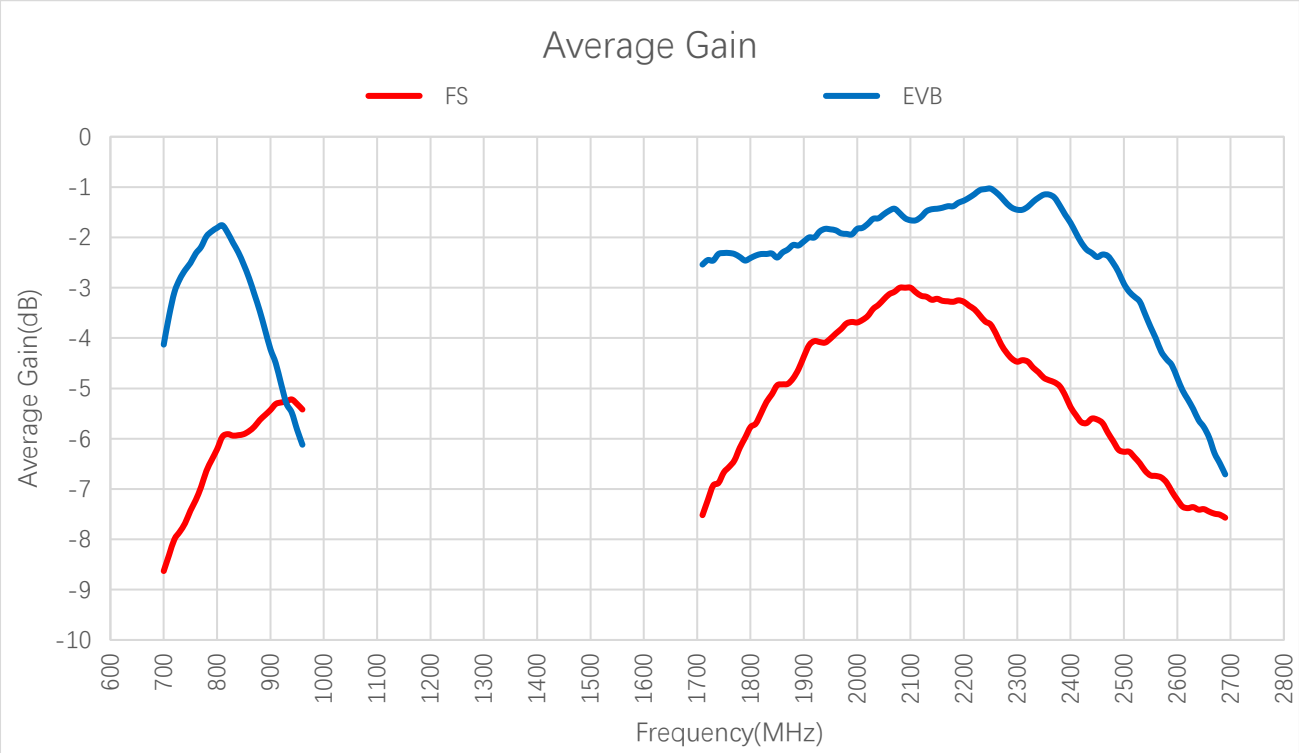
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	14.7	25.5	28.7	28.7	-	17.7	20.5	33.1
EVB	-	-	44.0	61.5	37.8	24.4	-	55.7	58.5	61.0
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	39.8	47.4	33.2	27.4	19.0	17.5	-	-	-	-
EVB	65.5	71.8	76.8	57.7	33.2	21.3	-	-	-	-

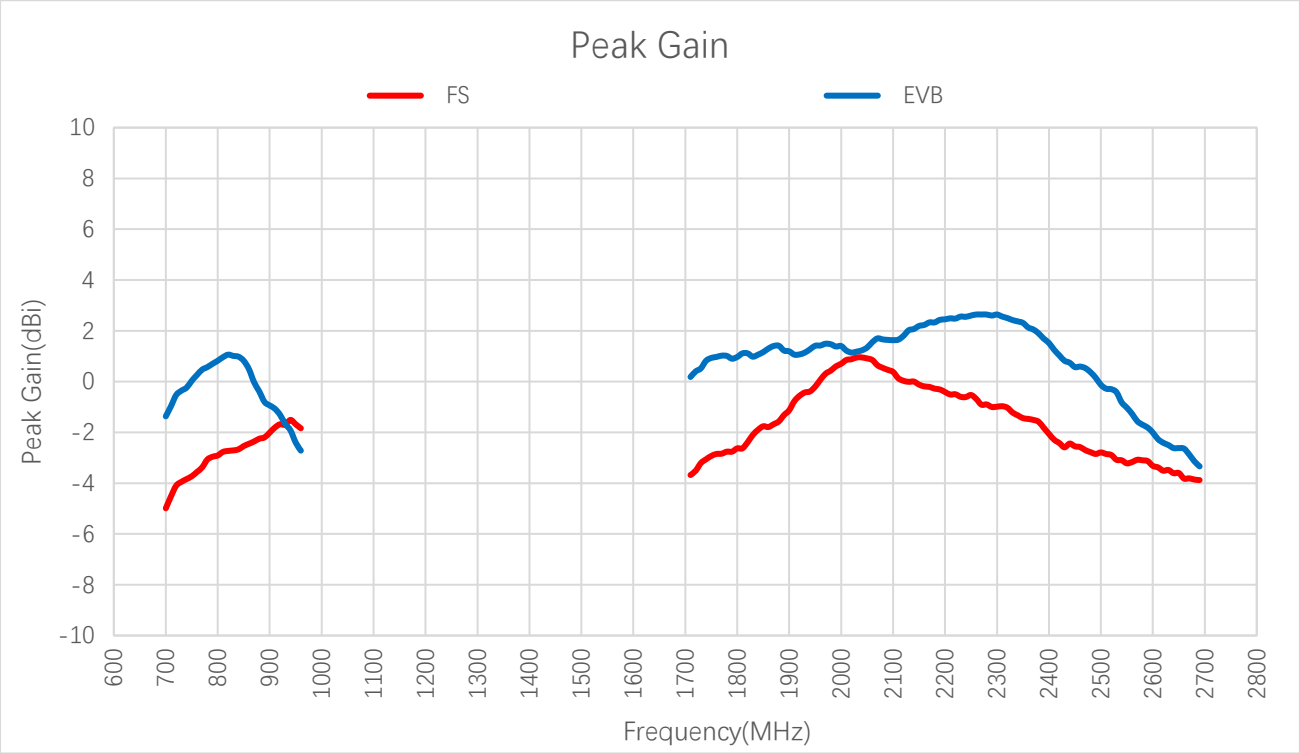
3.2.2. Average Gain



Average Gain (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-8.3	-5.9	-5.4	-5.4	-	-7.5	-6.9	-4.8
EVB	-	-	-3.6	-2.1	-4.2	-6.1	-	-2.5	-2.3	-2.2
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-4.0	-3.2	-4.8	-5.6	-7.2	-7.6	-	-	-	-
EVB	-1.8	-1.4	-1.2	-2.4	-4.8	-6.7	-	-	-	-

3.2.3. Peak Gain



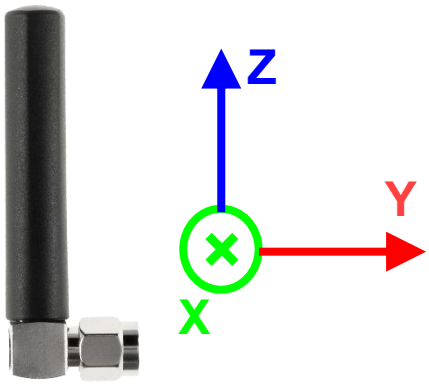
Peak Gain (dBi)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-4.5	-2.7	-2.0	-1.8	-	-3.7	-3.1	-1.6
EVB	-	-	-1.0	1.0	-0.9	-2.7	-	0.2	0.8	1.4
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-0.2	0.0	-1.4	-2.6	-3.3	-3.9	-	-	-	-
EVB	1.4	2.1	2.3	0.6	-2.0	-3.3	-	-	-	-

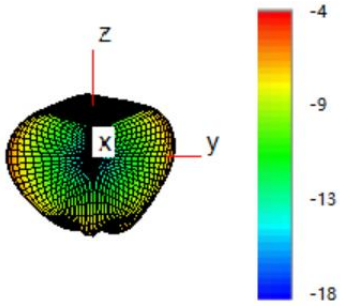
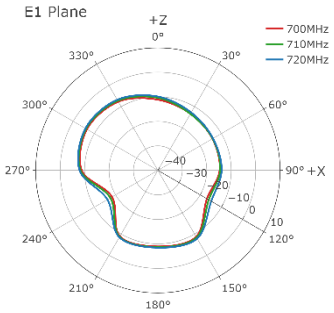
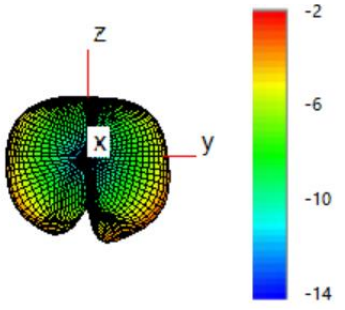
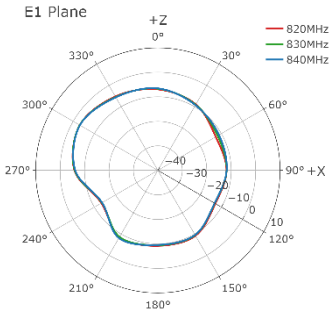
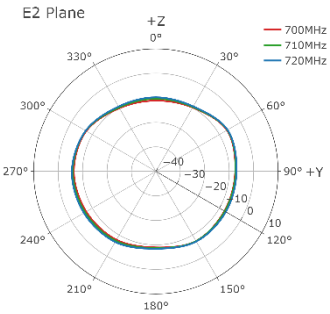
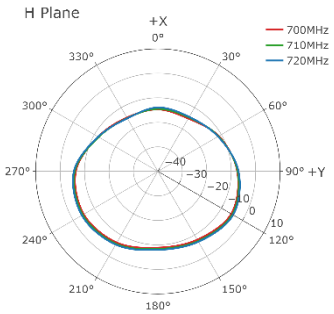
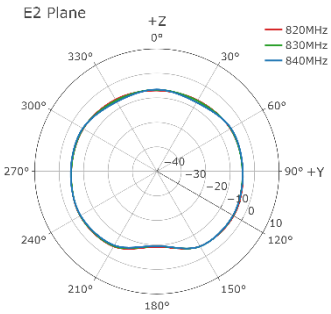
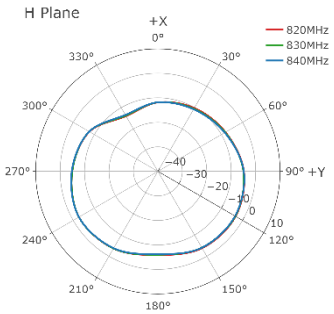
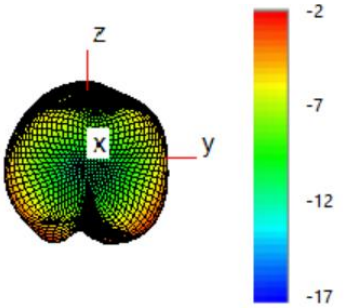
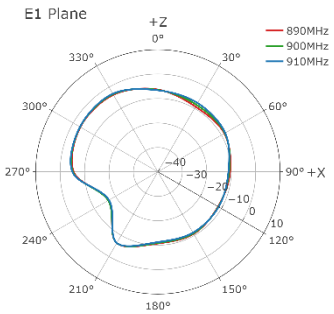
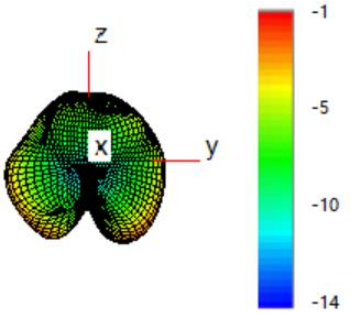
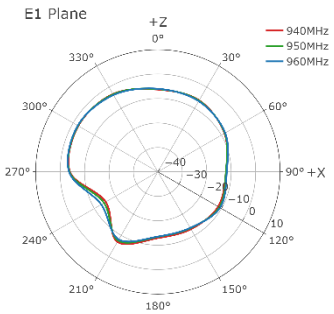
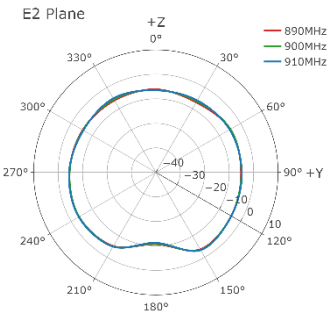
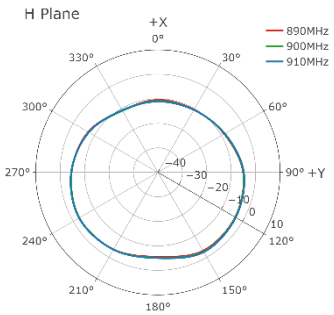
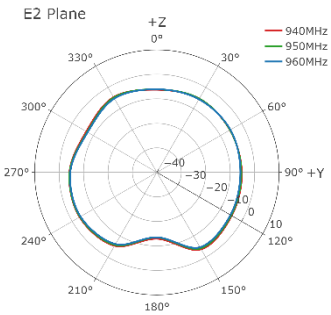
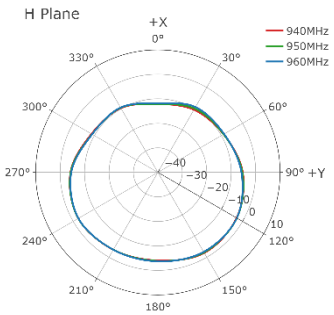
3.2.4. 3D & 2D Radiation Pattern

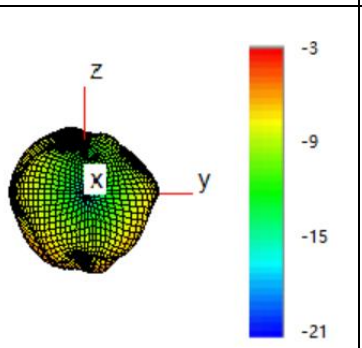
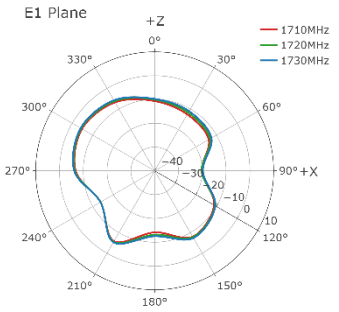
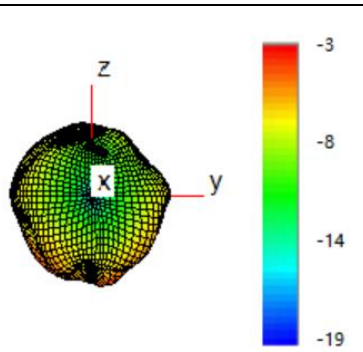
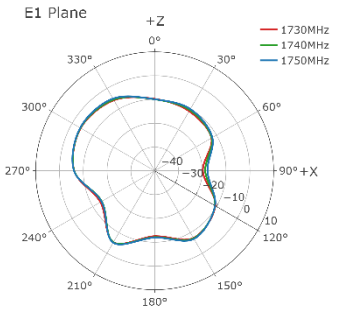
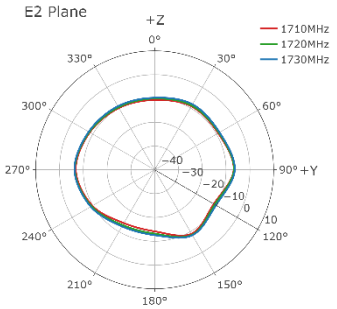
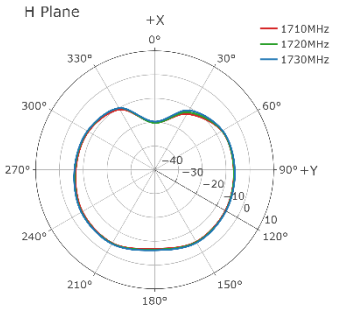
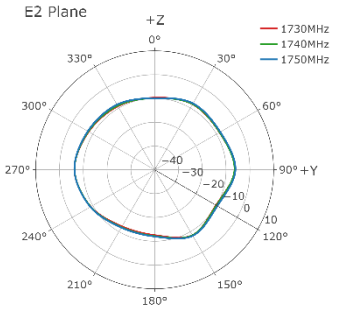
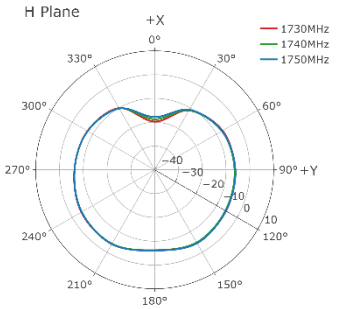
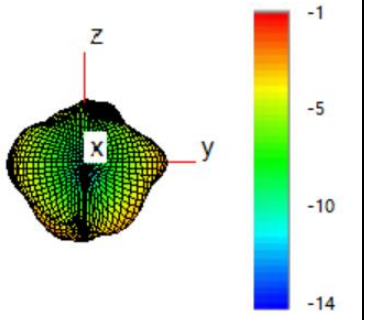
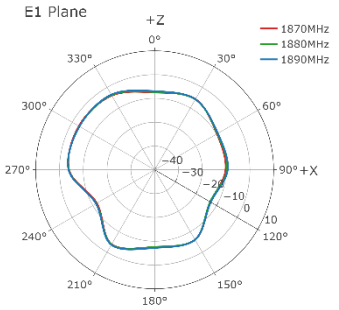
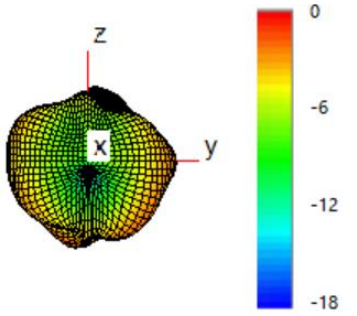
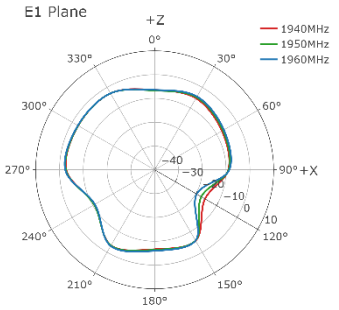
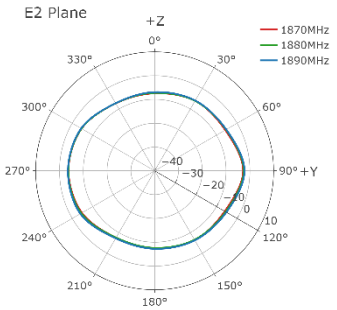
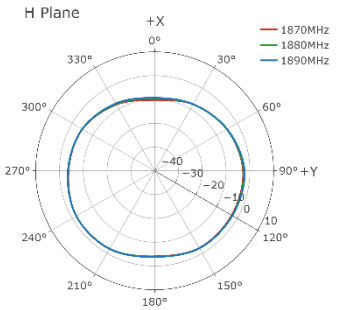
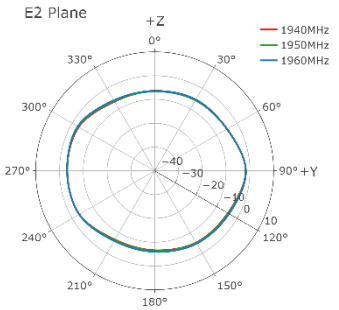
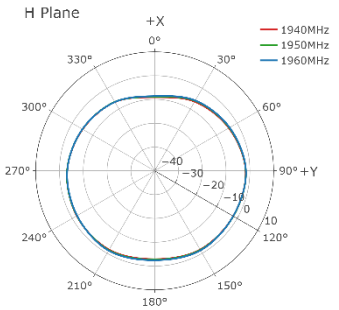
3.2.4.1. Test Condition: In Free Space

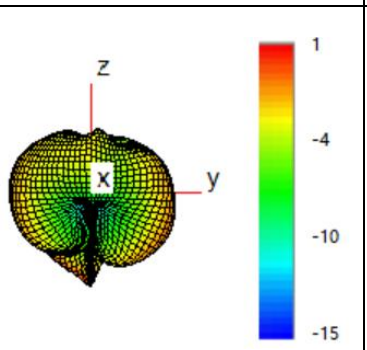
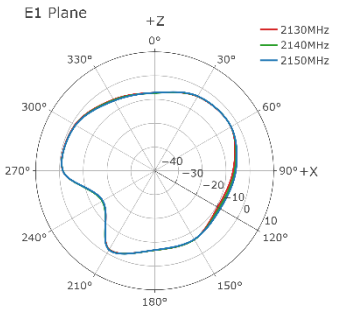
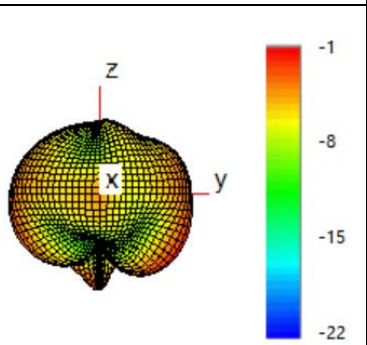
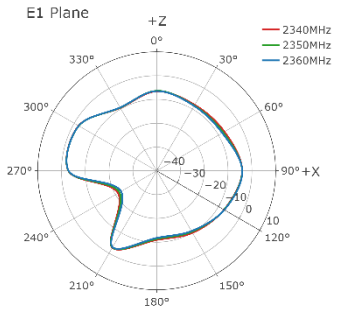
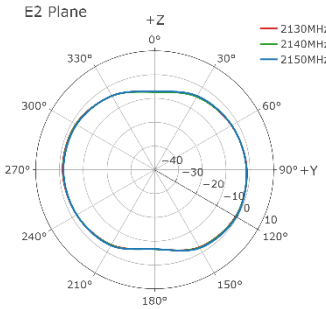
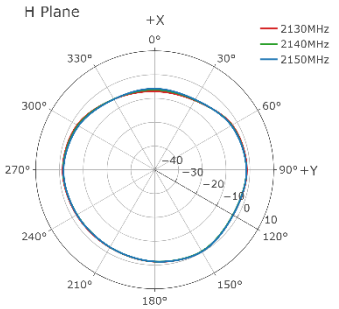
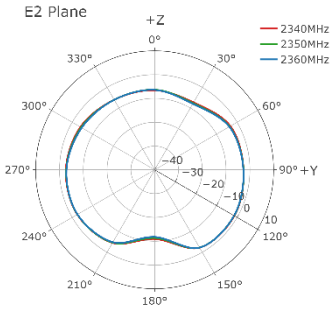
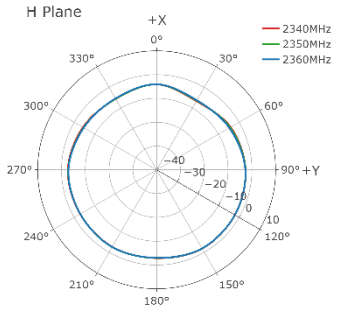
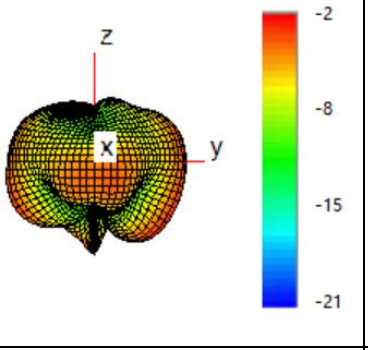
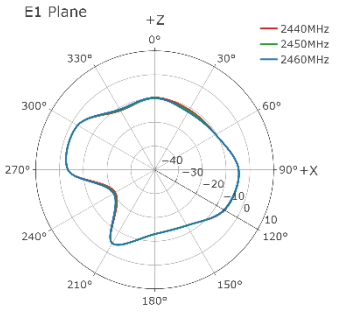
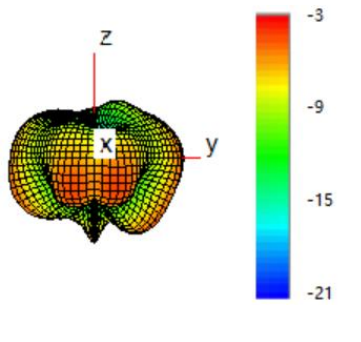
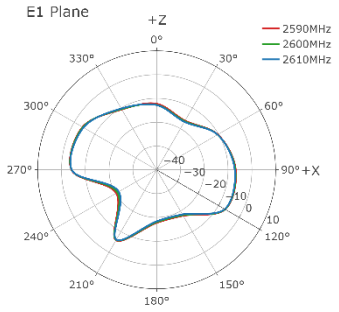
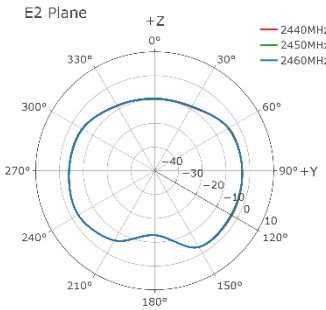
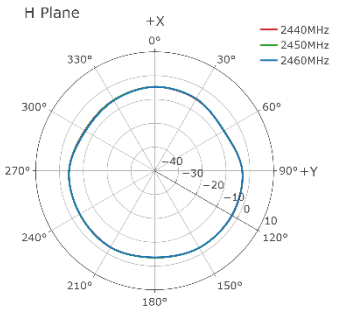
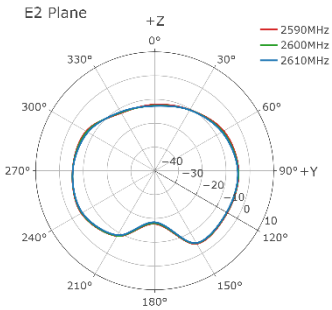
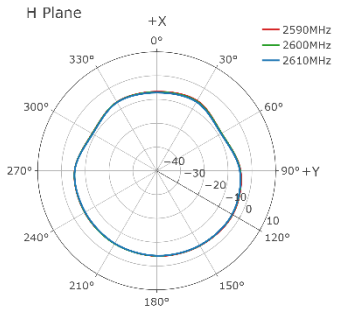
- Test Chamber: HF-G-1

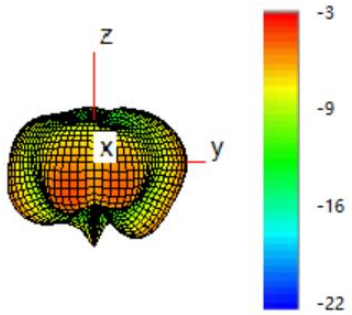
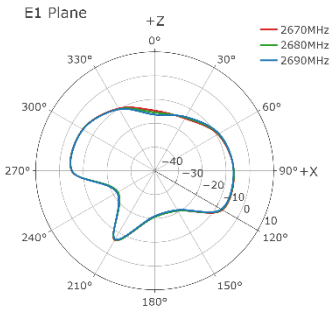
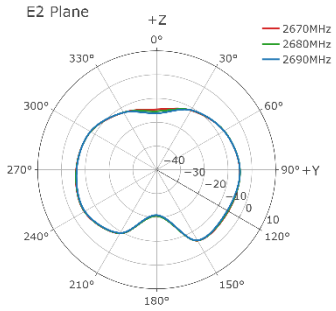
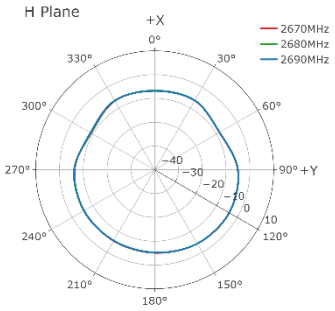


- **FS**

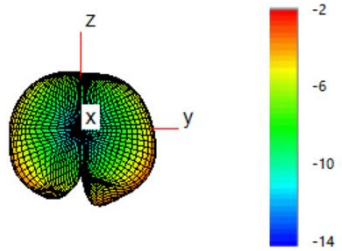
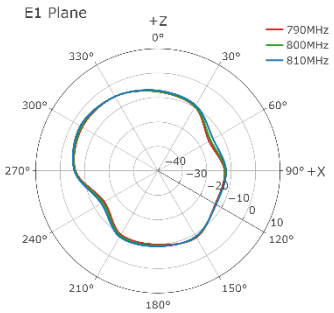
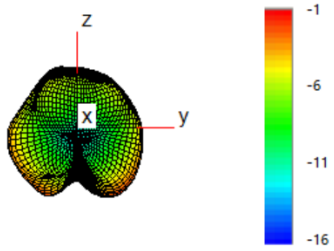
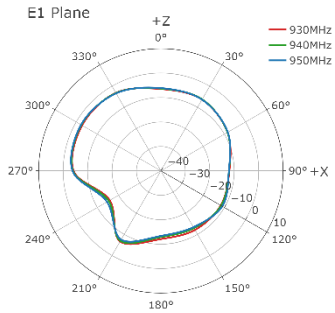
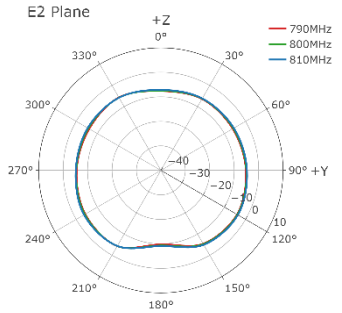
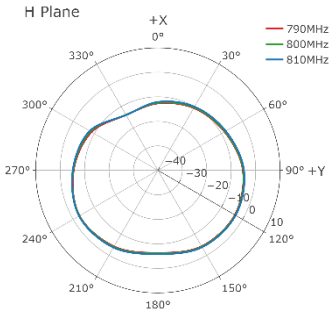
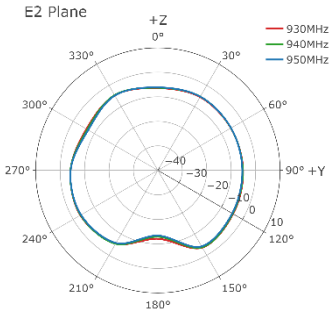
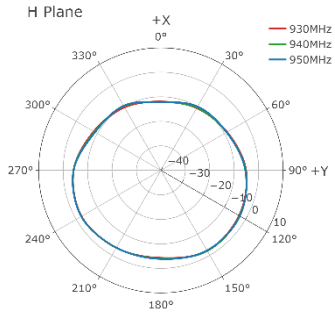
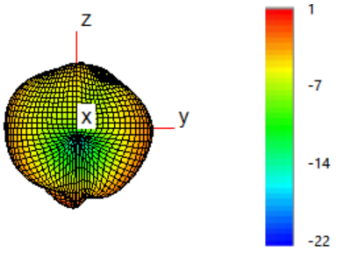
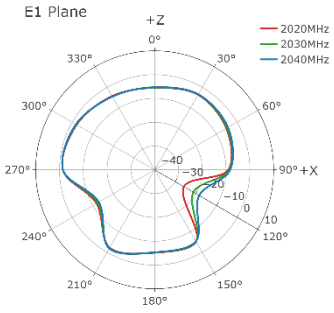
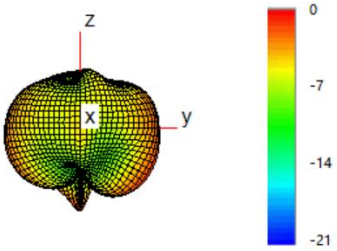
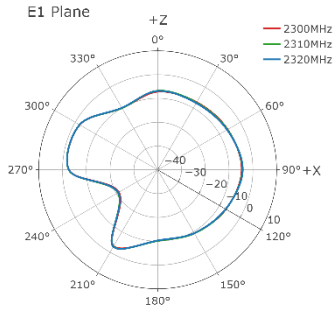
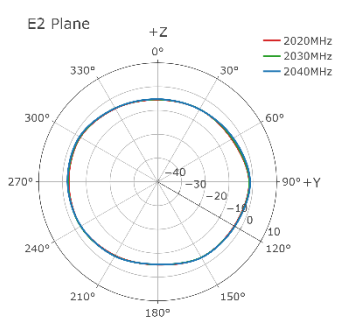
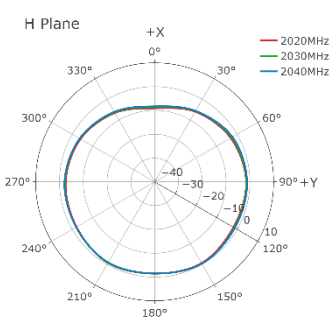
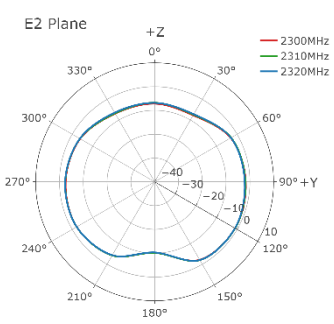
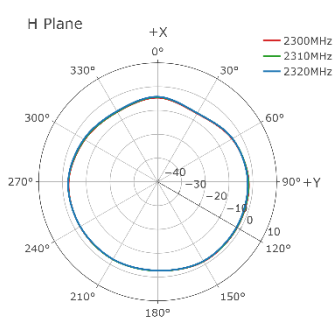
710 MHz		830 MHz	
			
			
900 MHz		960 MHz	
			
			

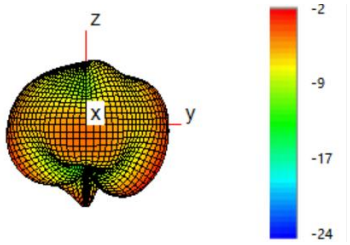
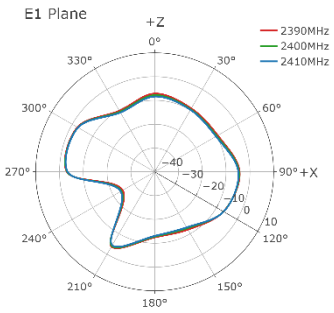
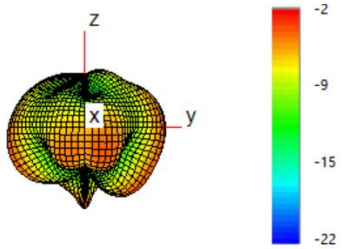
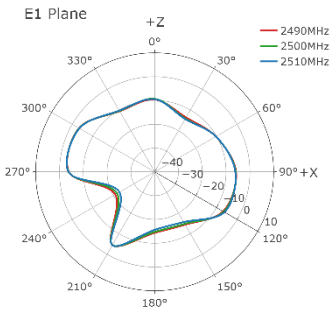
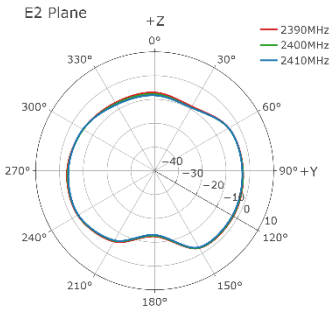
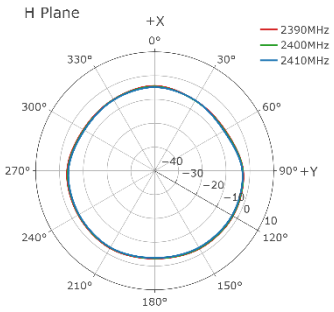
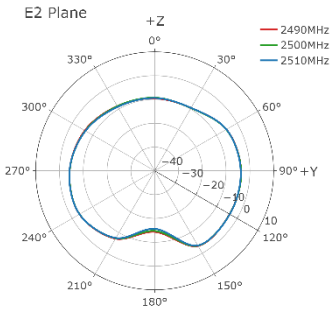
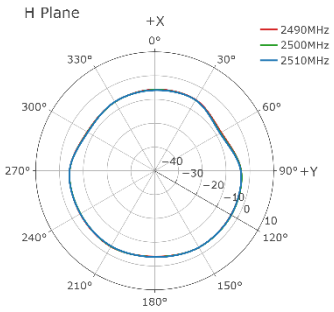
1710 MHz		1740 MHz	
			
			
1880 MHz		1950 MHz	
			
			

2140 MHz		2350 MHz	
			
			
2450 MHz		2600 MHz	
			
			

2690 MHz	
 z x y -3 -9 -16 -22	 E1 Plane +Z 2670MHz 2680MHz 2690MHz 0° 30° 60° 90° +X 120° 150° 180° 210° 240° 270° 300° 330° -40 -30 -20 -10 0 10
 E2 Plane +Z 2670MHz 2680MHz 2690MHz 0° 30° 60° 90° +Y 120° 150° 180° 210° 240° 270° 300° 330° -40 -30 -20 -10 0 10	 H Plane +X 2670MHz 2680MHz 2690MHz 0° 30° 60° 90° +Y 120° 150° 180° 210° 240° 270° 300° 330° -40 -30 -20 -10 0 10

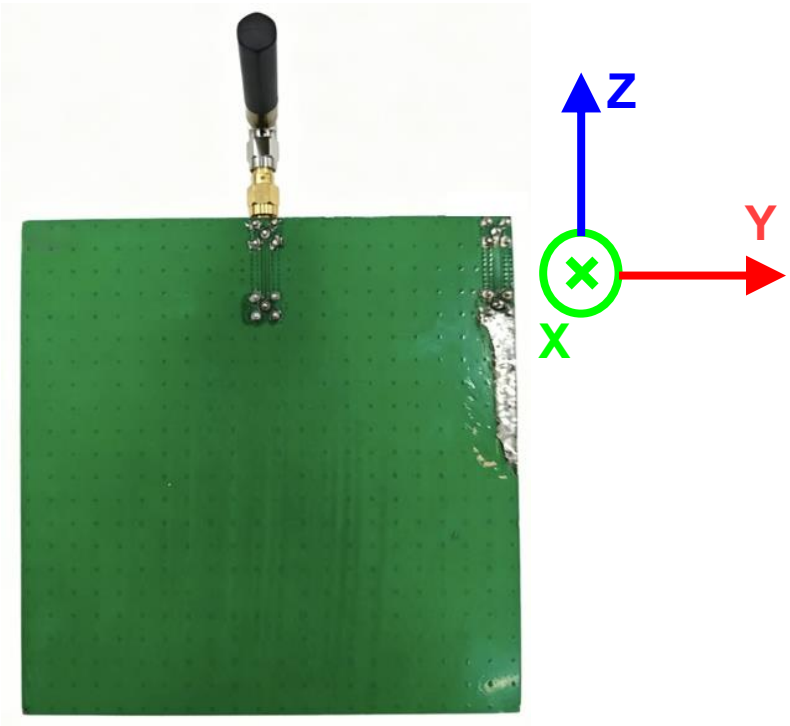
- **Max Peak Gain**

810 MHz		940 MHz	
			
			
2030 MHz		2310 MHz	
			
			

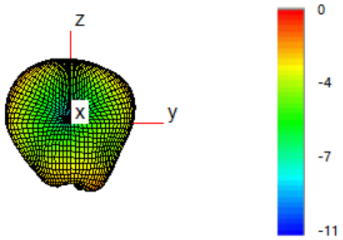
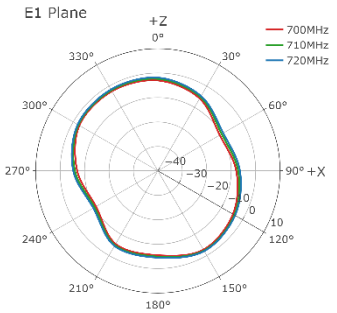
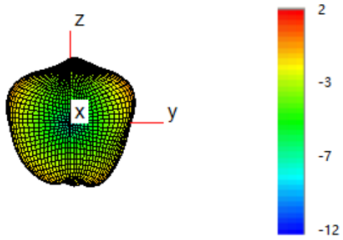
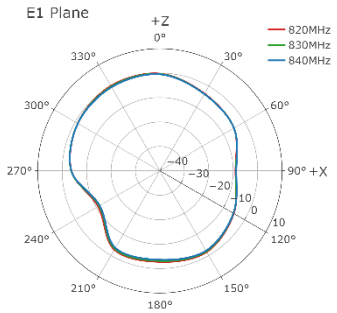
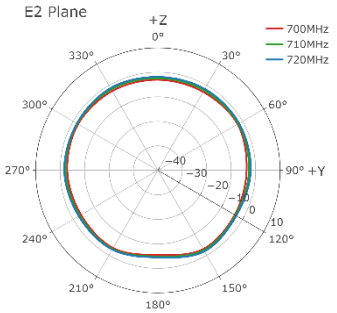
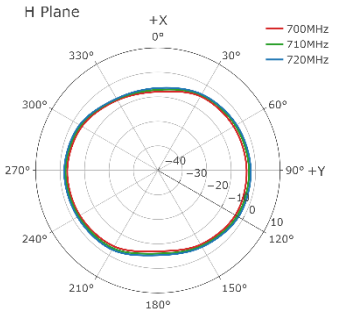
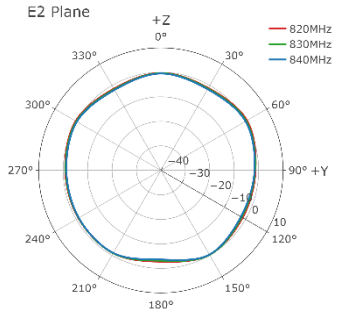
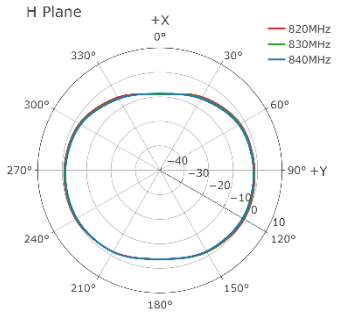
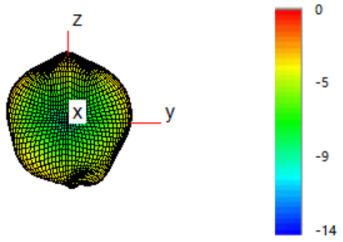
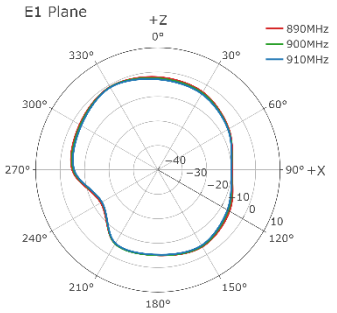
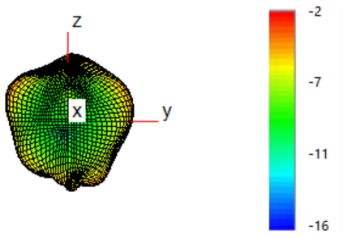
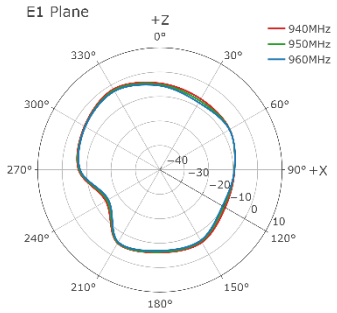
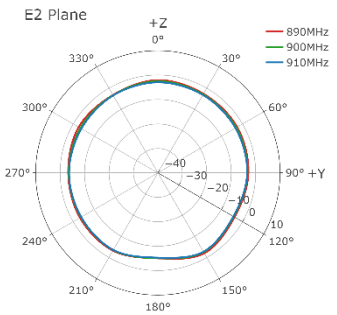
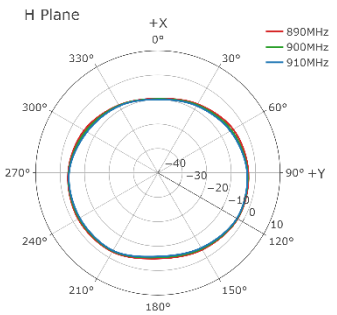
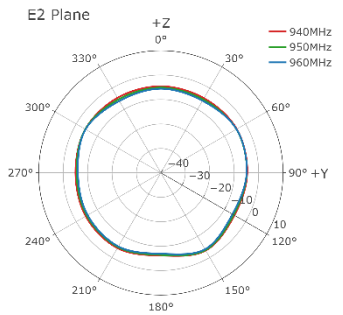
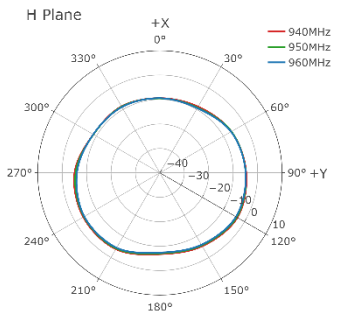
2400 MHz		2500 MHz	
			
			

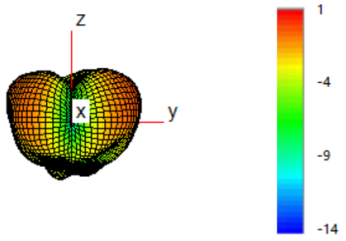
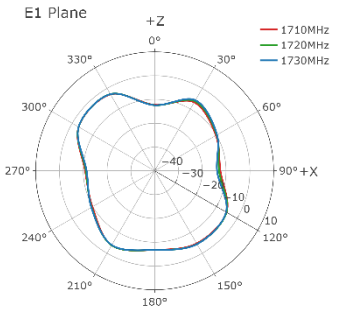
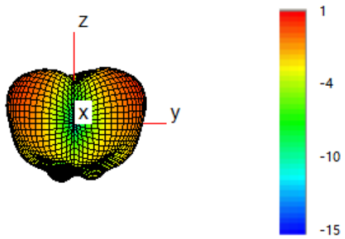
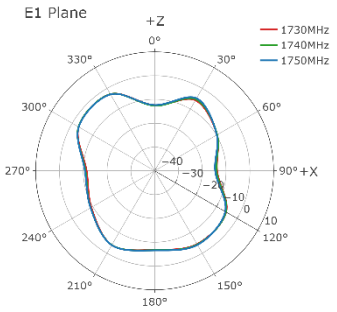
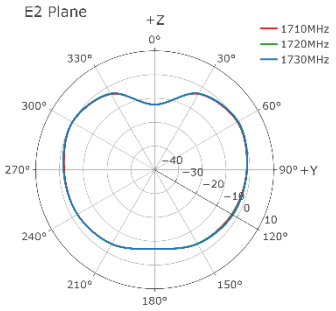
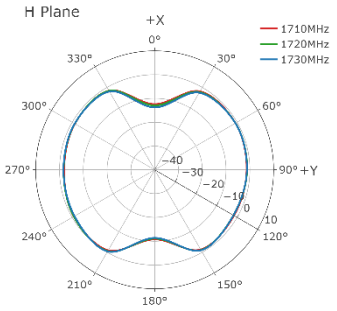
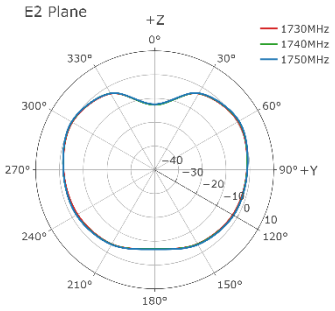
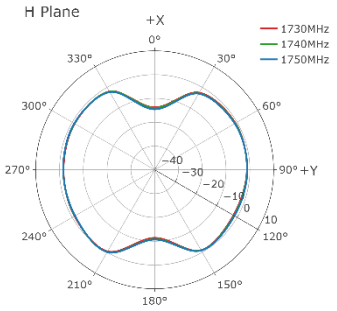
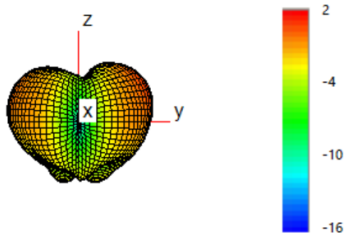
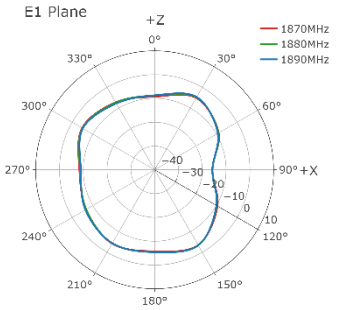
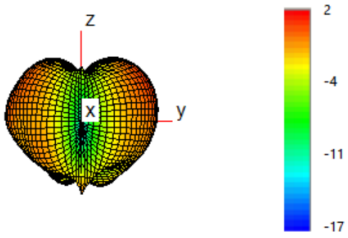
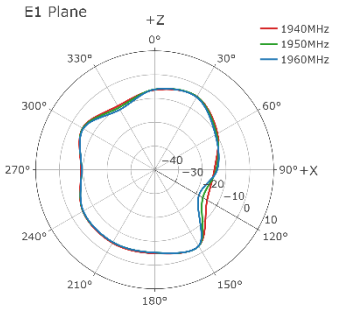
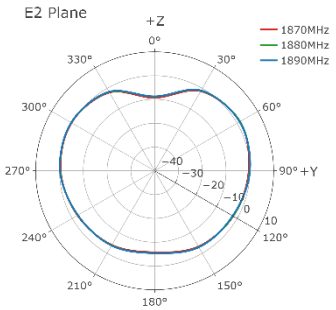
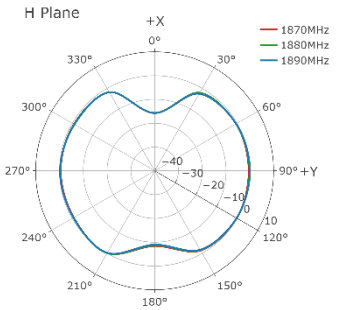
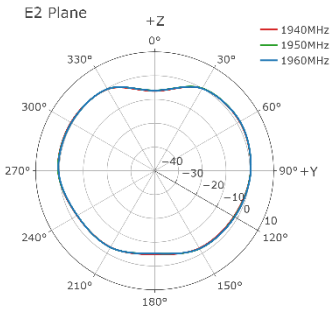
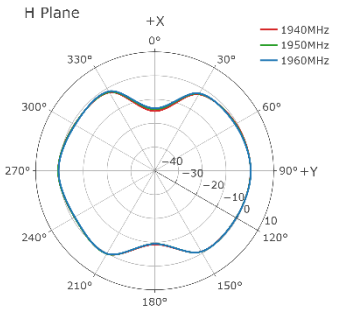
3.2.4.2. Test Condition: On 130 mm × 130 mm EVB

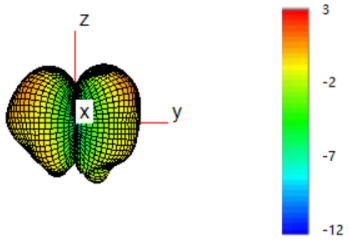
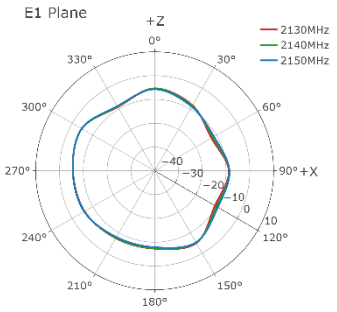
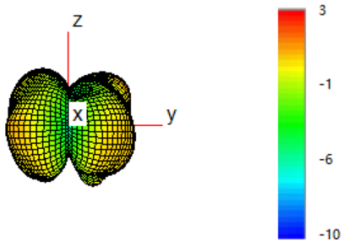
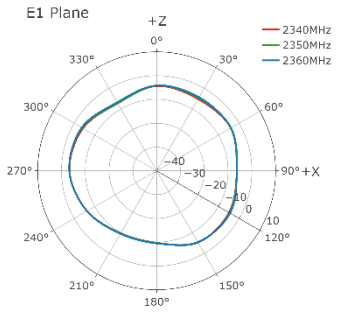
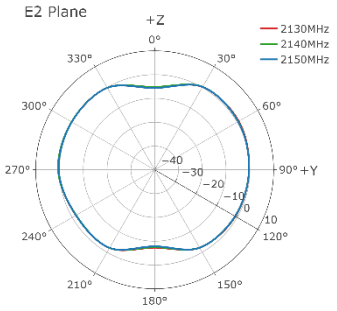
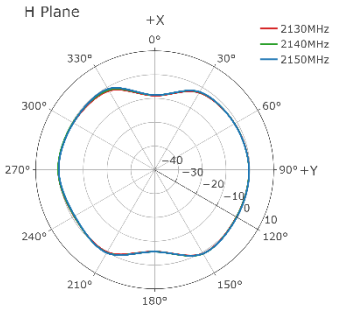
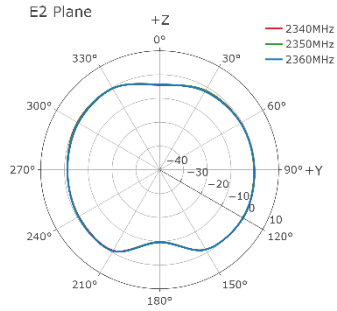
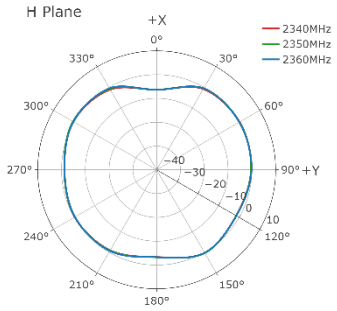
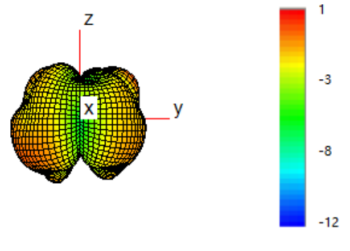
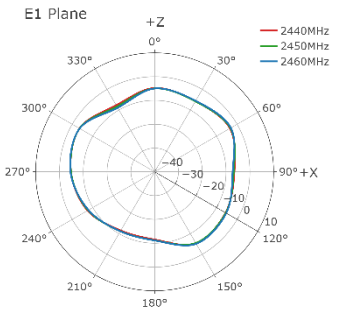
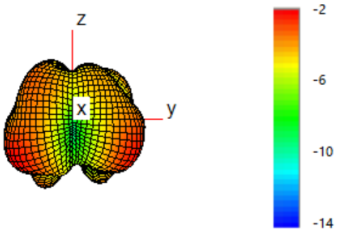
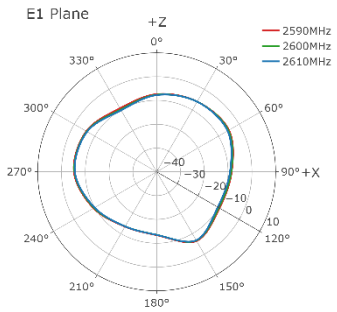
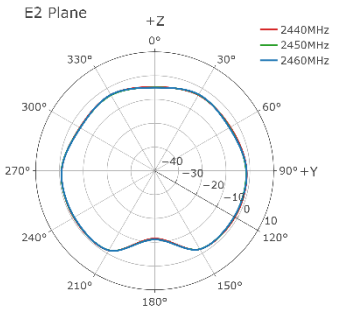
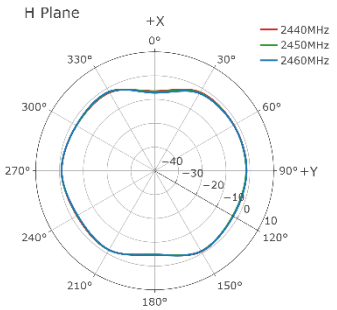
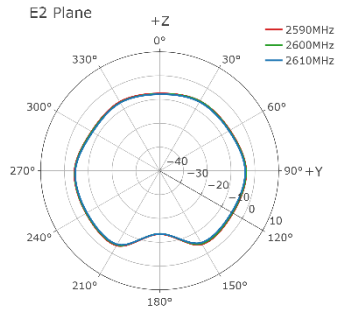
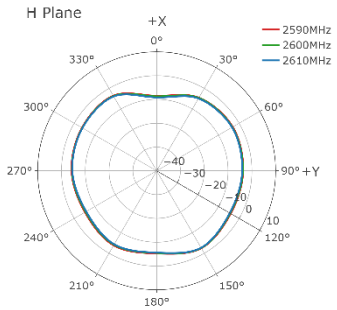
- Test Chamber: HF-G-1

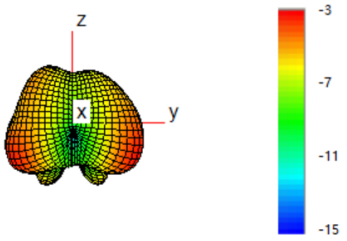
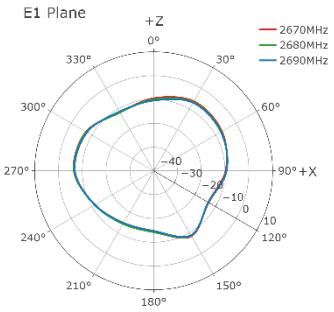
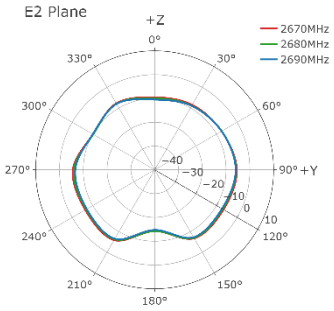
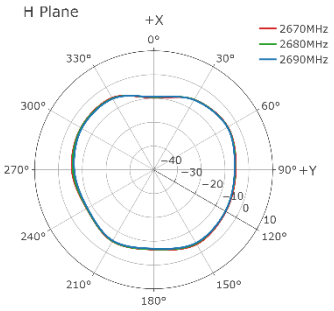


- **EVB**

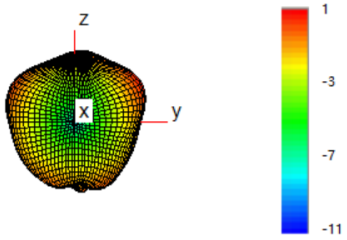
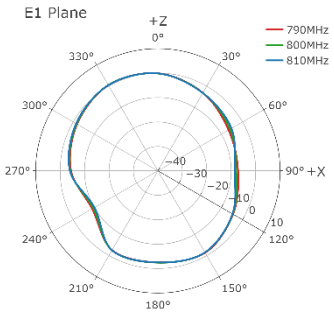
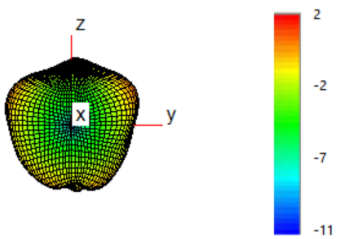
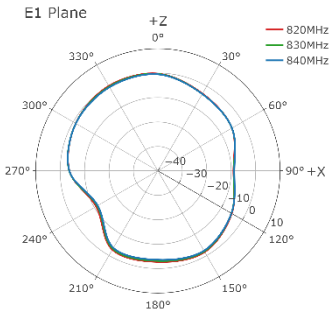
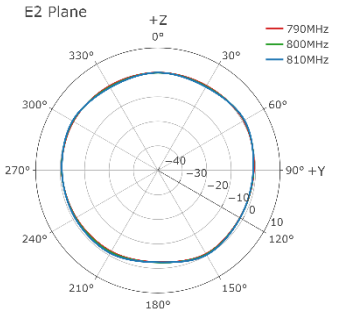
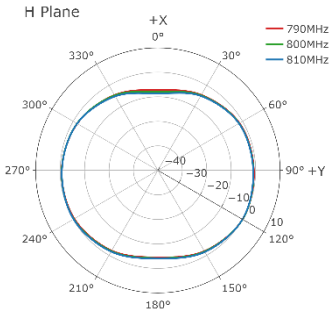
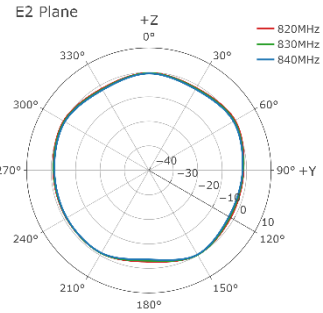
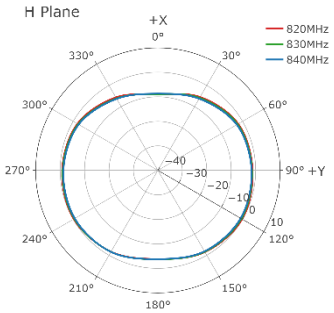
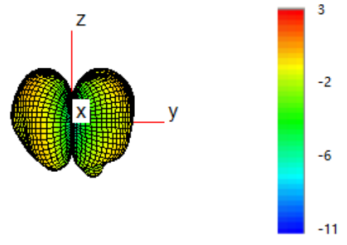
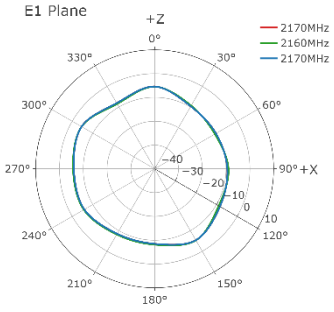
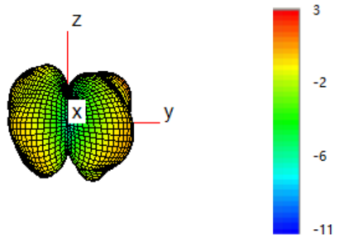
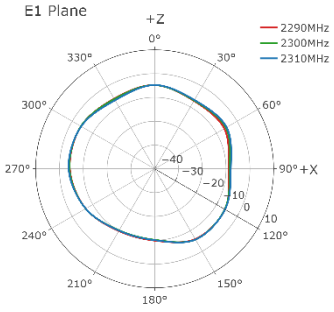
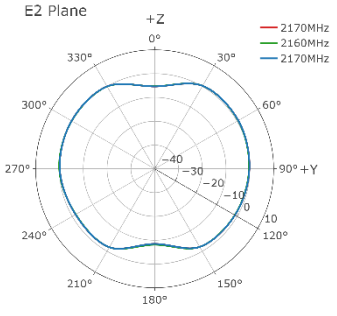
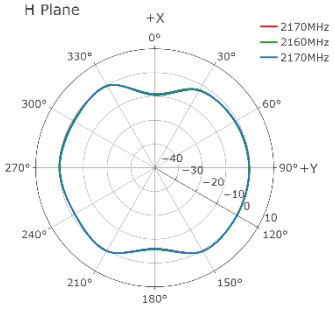
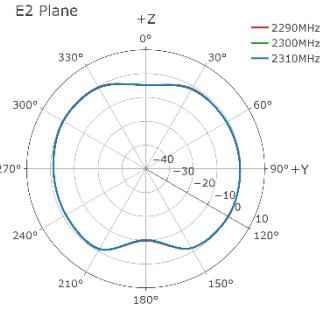
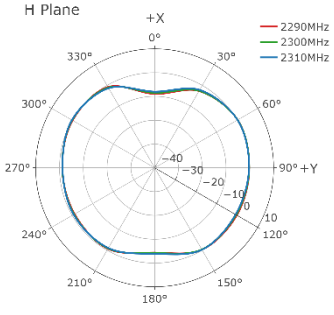
710 MHz		830 MHz	
			
			
900 MHz		960 MHz	
			
			

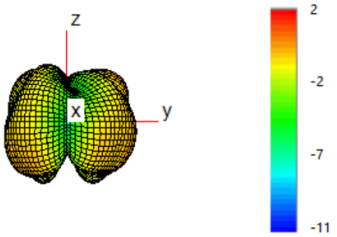
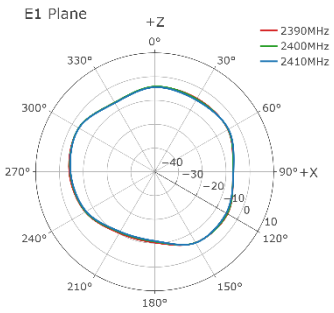
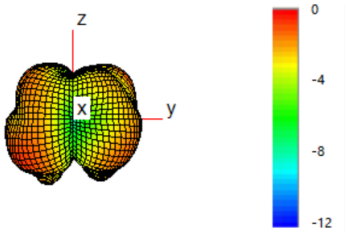
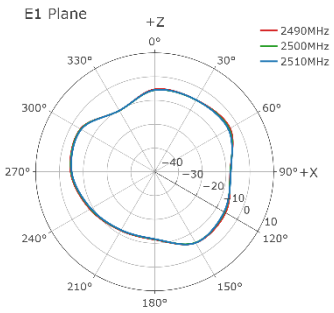
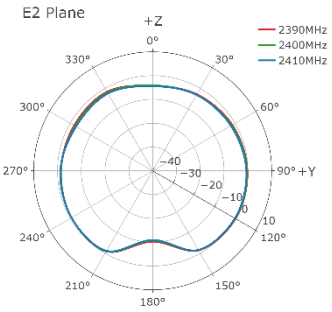
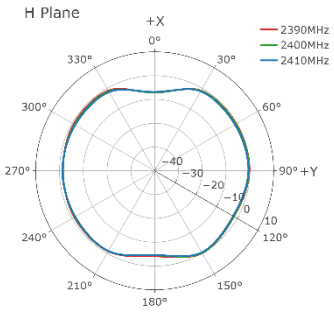
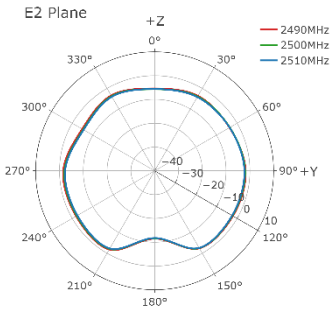
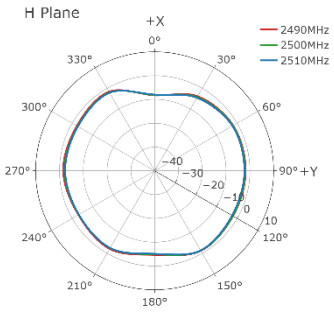
1710 MHz		1740 MHz	
			
			
1880 MHz		1950 MHz	
			
			

2140 MHz		2350 MHz	
			
			
2450 MHz		2600 MHz	
			
			

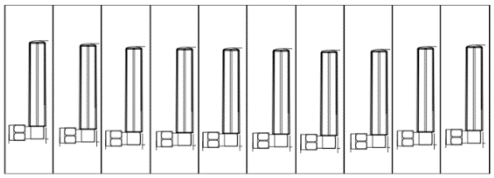
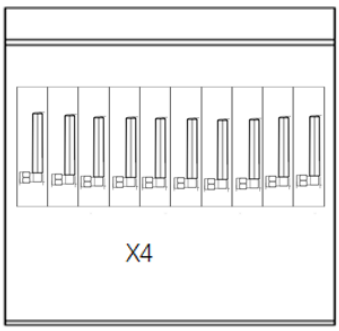
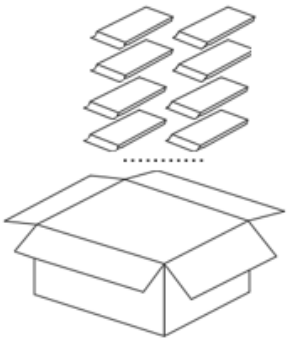
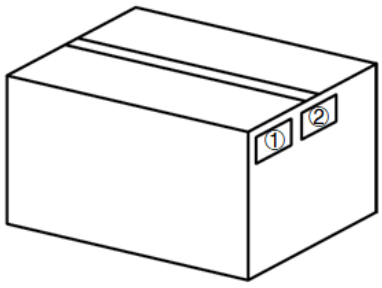
2690 MHz	
 Z X y -3 -7 -11 -15	 E1 Plane +Z 2670MHz 2680MHz 2690MHz 0° 30° 60° 90° +X 120° 150° 180° 210° 240° 270° 300° 330° -40 -30 -20 -10 0 10
 E2 Plane +Z 2670MHz 2680MHz 2690MHz 0° 30° 60° 90° +Y 120° 150° 180° 210° 240° 270° 300° 330° -40 -30 -20 -10 0 10	 H Plane +X 2670MHz 2680MHz 2690MHz 0° 30° 60° 90° +Y 120° 150° 180° 210° 240° 270° 300° 330° -40 -30 -20 -10 0 10

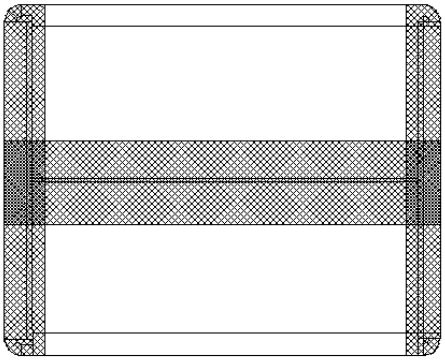
- **Max Peak Gain**

810 MHz		820 MHz	
			
			
2170 MHz		2300 MHz	
			
			

2400 MHz		2500 MHz	
			
			

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		10 antenna products in a one-piece bag. (10 Antennas / One-piece Bag)
2		40 antenna products in a PE bag. (40 Antennas / PE Bag)
3		(40 PE Bags / Carton Box) (1600 Antennas / Carton Box) Estimated quantity Products that cannot fill the entire carton box are packed in a suitable size carton box. Carton Size: L × W × H = 470 × 310 × 310 mm
4		**Position for Attaching Labels** ① Carton Label ② Quality Label

5		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Or our local offices. For more information, please visit:
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Revision History

Version	Date	Author	Note
-	2026-07-08	Ruean Liu/ Steven Mo/ Piers Ji/ Rainey Liao	Creation of the document
1.0	2026-07-08	Ruean Liu/ Steven Mo/ Piers Ji/ Rainey Liao	First official release



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